SCION: The Structural Intelligence Engine

Transitioning Teradata DNA from Rented Capability to Owned Intelligence.

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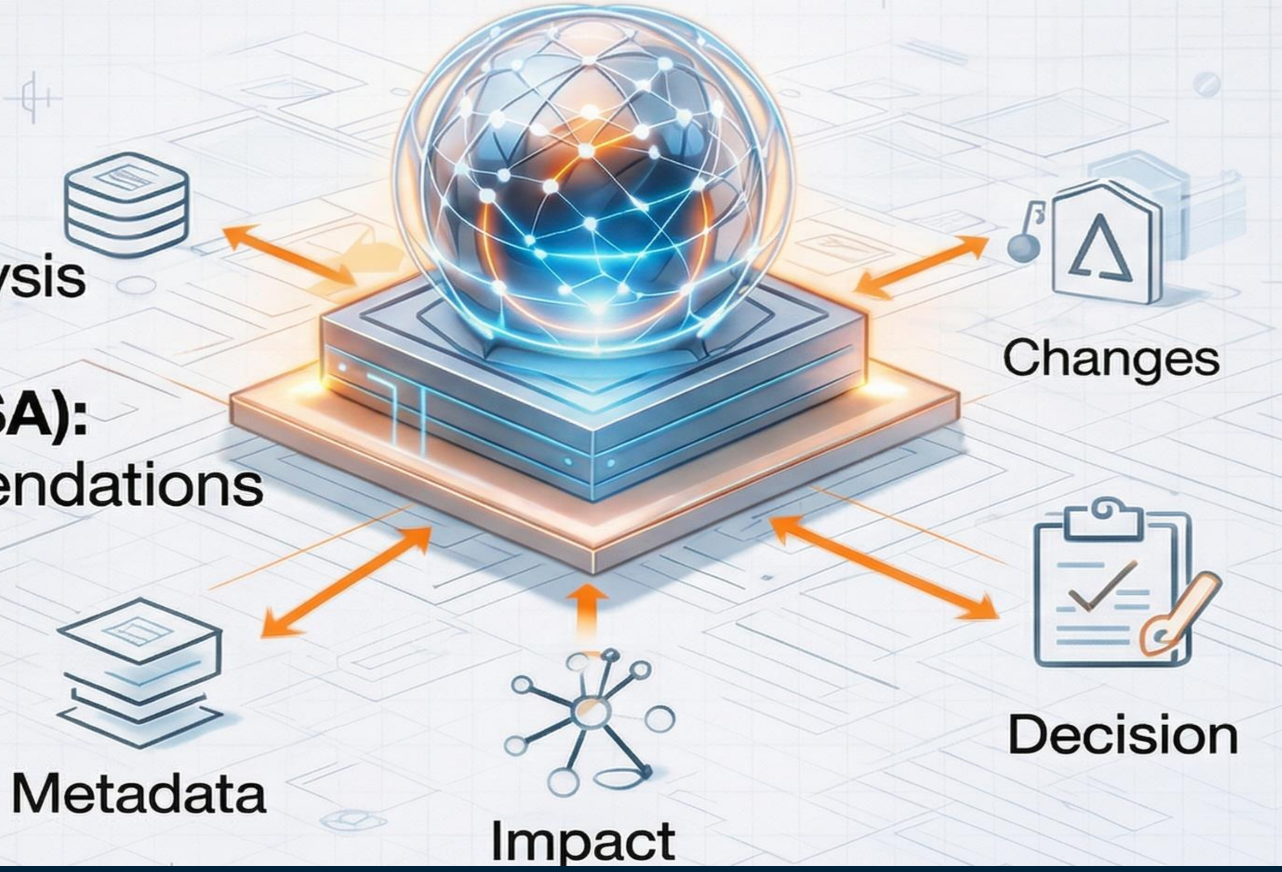
AGENDA

- **SCION** Overview
- Scope & Architecture
- **Estimation** *(Resources, Infrastructure, Maintenance)*
- Consolidated Summary
- Final Recommendations

What is SCION?

SCION is a Structural Intelligence Engine that transforms raw metadata into explainable, actionable technical decisions.

- **Snapshot + Diff + Graph** for metadata versioning, change detection, & impact analysis
- **Explainable AI reasoning (TAISA):** for risk evaluation and recommendations
- **Full traceability and control** over technical evolution.



End-to-End Flow: Parser → SCION → Insight

Parser
(External - DataDNA)

- ***Extracts*** structured metadata from source systems
- ***Processes*** SQL / scripts (BTEQ, etc.)
- ***Outputs*** structured metadata (JSON / Excel)
- ***Stateless*** (no state stored)

Produces structured metadata (stateless)

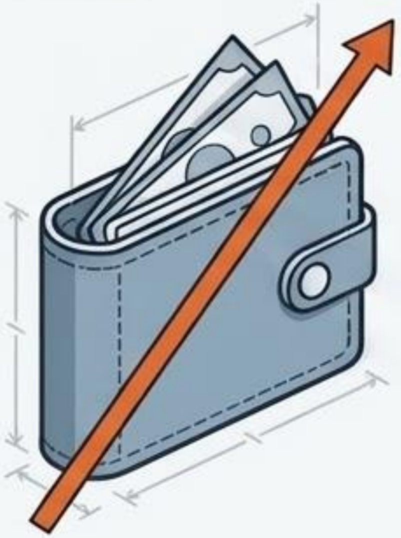
SCION
(Structural Intelligence Engine)

- ***Stores*** versioned metadata (Snapshots)
- ***Detects*** structural changes (Diff)
- ***Builds*** lineage & impact graph (Graph)
- ***Intelligent*** stateful core

Aligned with **DataDNA** architecture: Parser performs extraction → **SCION** stores and analyzes

The Case for Proprietary Governance

Key Metric



Eliminate Cost

Dramatic Monthly Spending Reduction

OPEX Reduction

Key Metric

Speed to Value

5–6 Months to complete implementation

Critical Path Analysis

Key Metric

Asset Value

0% → 100% Proprietary IP Ownership

No Third-Party Dependency

EXECUTIVE TAKEAWAYS

The Mandate: Replace opaque ‘black-box” legacy tools to eliminate exposure and recurring fees.

The Solution: Deploy SCION—a deterministic, modular governance platform with native AI augmentation.

Architecture Overview: A Layered, Modular Design

- **Ingestion:** Parser-driven, stateless lineage derivation (JSON / Excel inputs).
- **Core Processing:** Deterministic state versioning and change detection via snapshots.
- **Impact Engine:** Applies parser-derived lineage using recursive graph logic.
- **Intelligence:** Usage-based weighting and volatility metrics.
- **TAISA AI:** Stateless insight generation from structured system events.
- **Delivery:** Dockerized runtime with secure API orchestration.

Layer 6: Delivery
(No engine, API & UI)

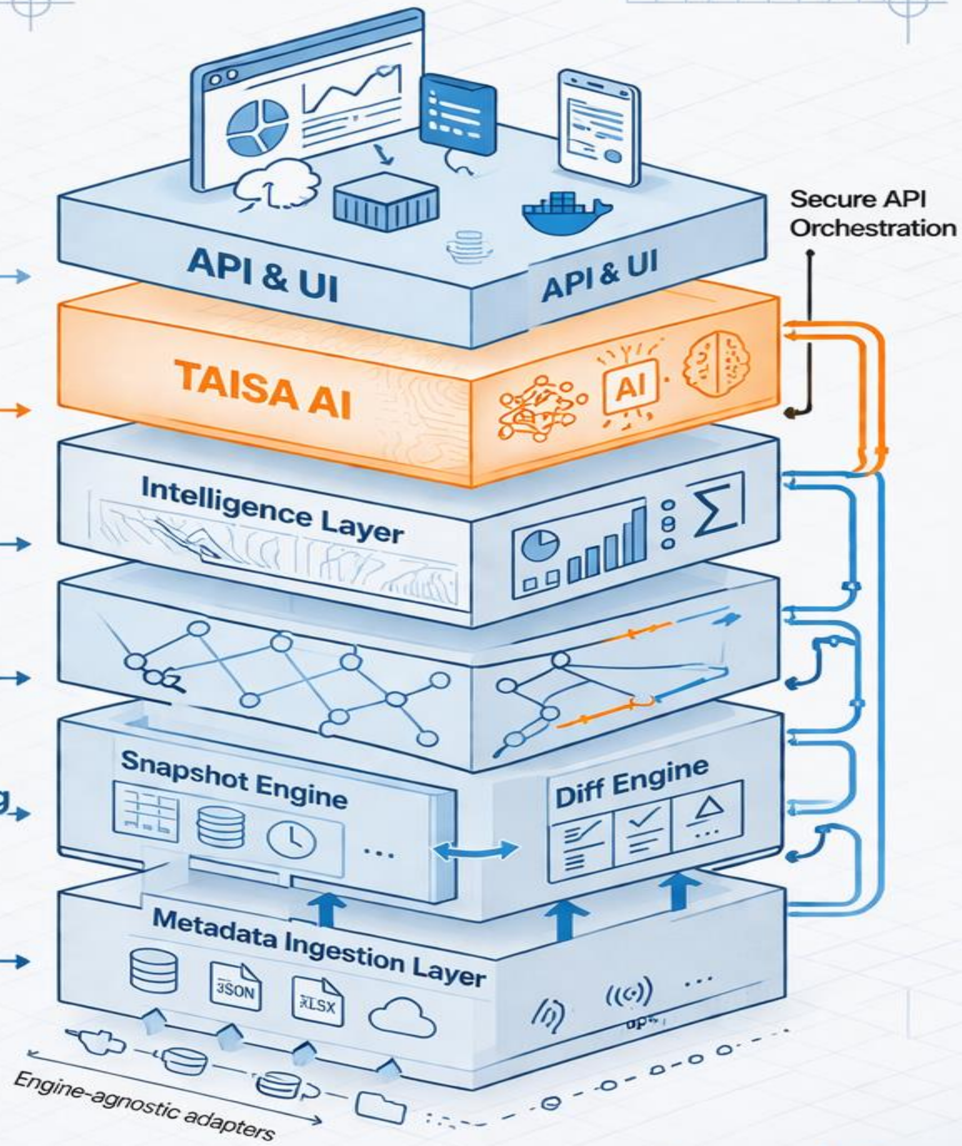
Layer 5: TAISA
Engine 7

Layer 4: Intelligence
Engine 5 (Usage)
Engine 6 (Metrics)

Layer 3: Impact
Engine 4 (Graph)

Layer 2: Core Processing
Engine 2 (Snapshot)
Engine 3 (Diff)

Layer 1: Ingestion
Engine 1



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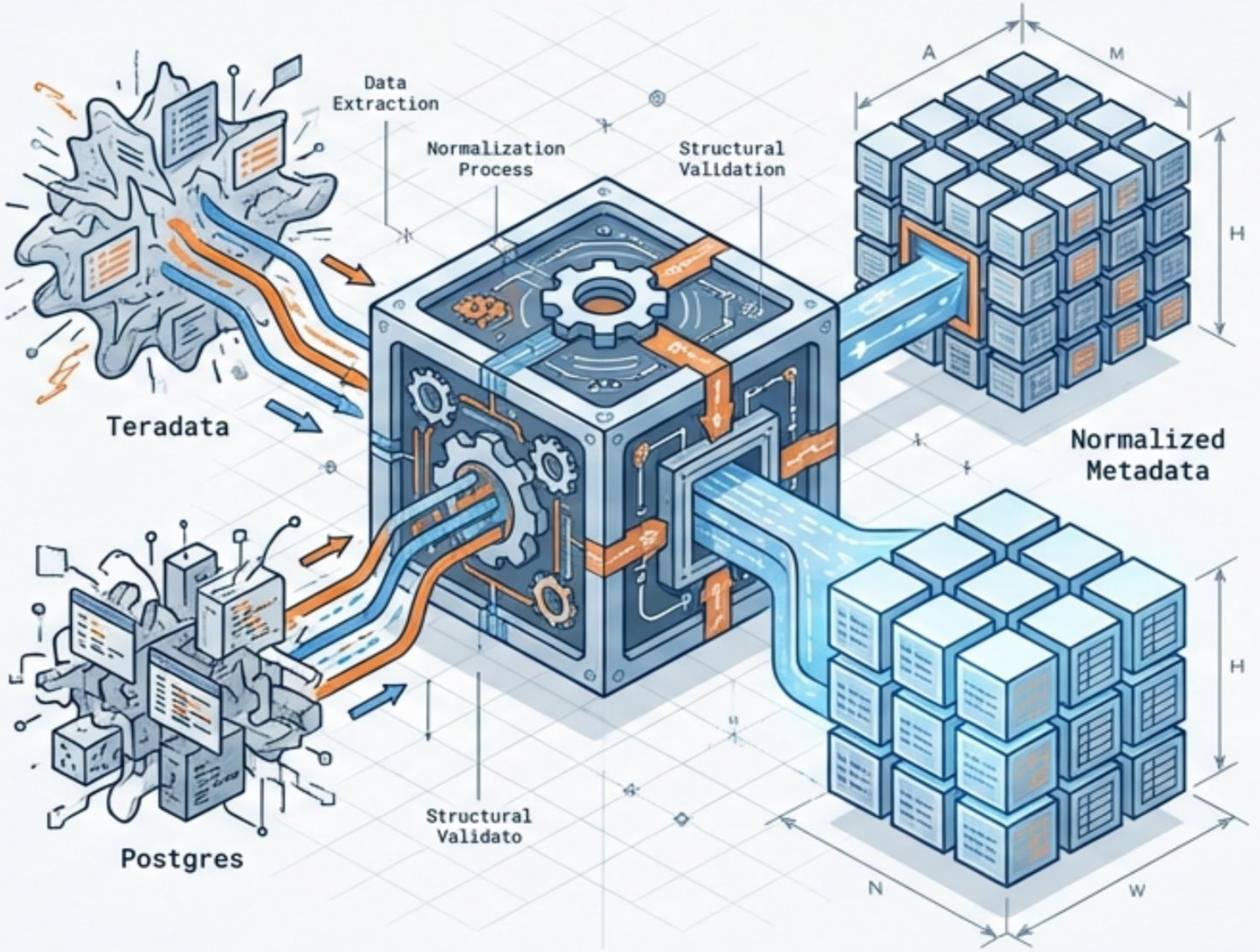
Engine 1: Metadata Ingestion Layer

Role / Purpose:
The Universal Adapter. Responsible for extracting and normalizing structural metadata from EDW engines.

Core Capabilities:
- Source-agnostic extraction (**Teradata/Postgres** adapters).
- Canonical **YAML**-based **template engine.**
- **Structural validation** and **typological normalization.**

Technical Value:
Decouples governance from specific database vendors; eliminates raw catalog replication.

Strategic Value:
Future-proofs the platform, allowing DNA to adapt to new EDW technologies without re-platforming.



Engine 2: Snapshot Engine

Role / Purpose:

The Time Machine. Captures full EDW state to guarantee transactional consistency.

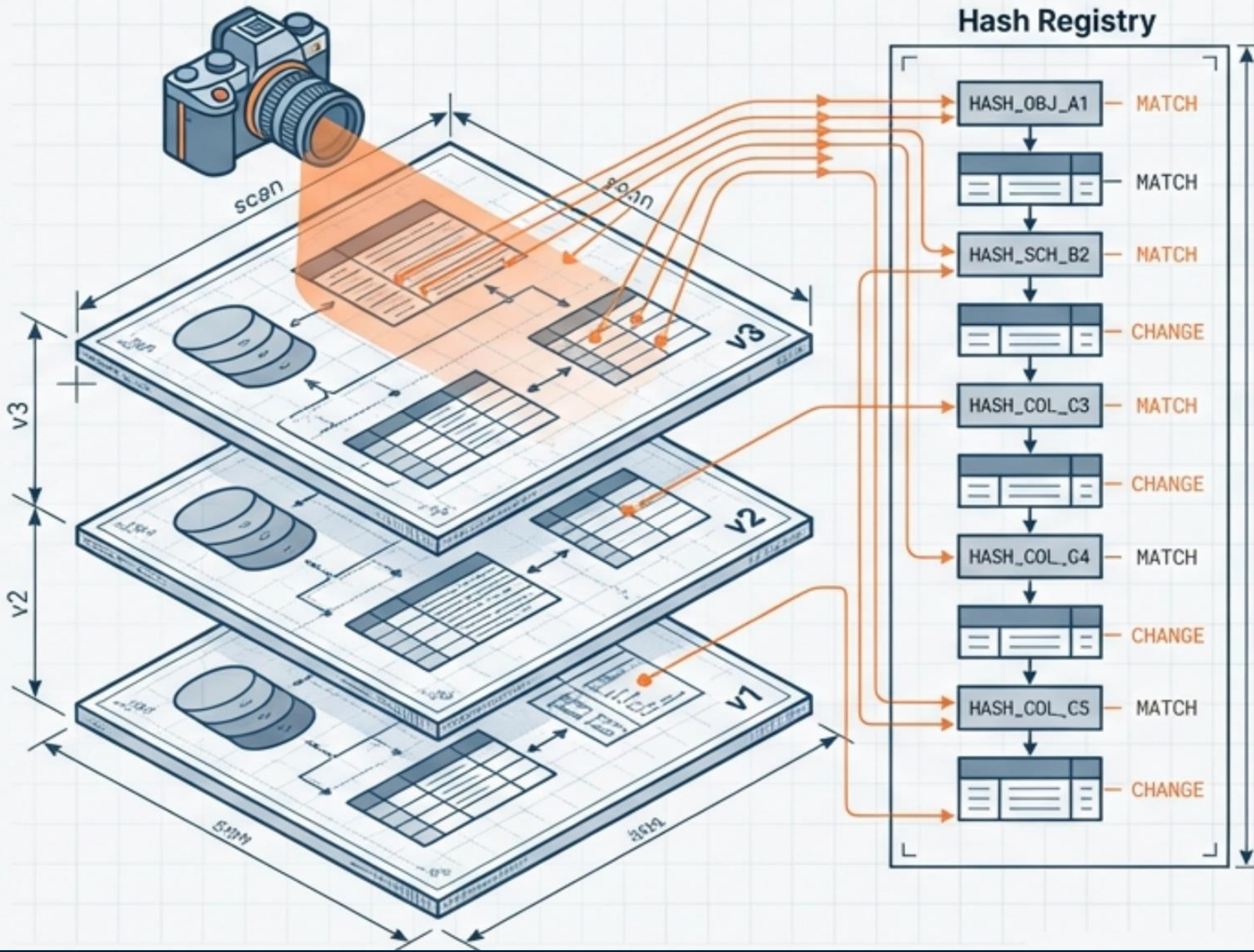
Core Capabilities:

- Historical versioning with structural hashing.
- Detects changes via hash comparison (Zero data read).
- Metrics capture: Object, Schema, and Column counts.

Technical Value: Moves from passive tracking to active state monitoring.

Strategic Value:

Provides the foundational ‘truth’ required for regulatory compliance and auditability.



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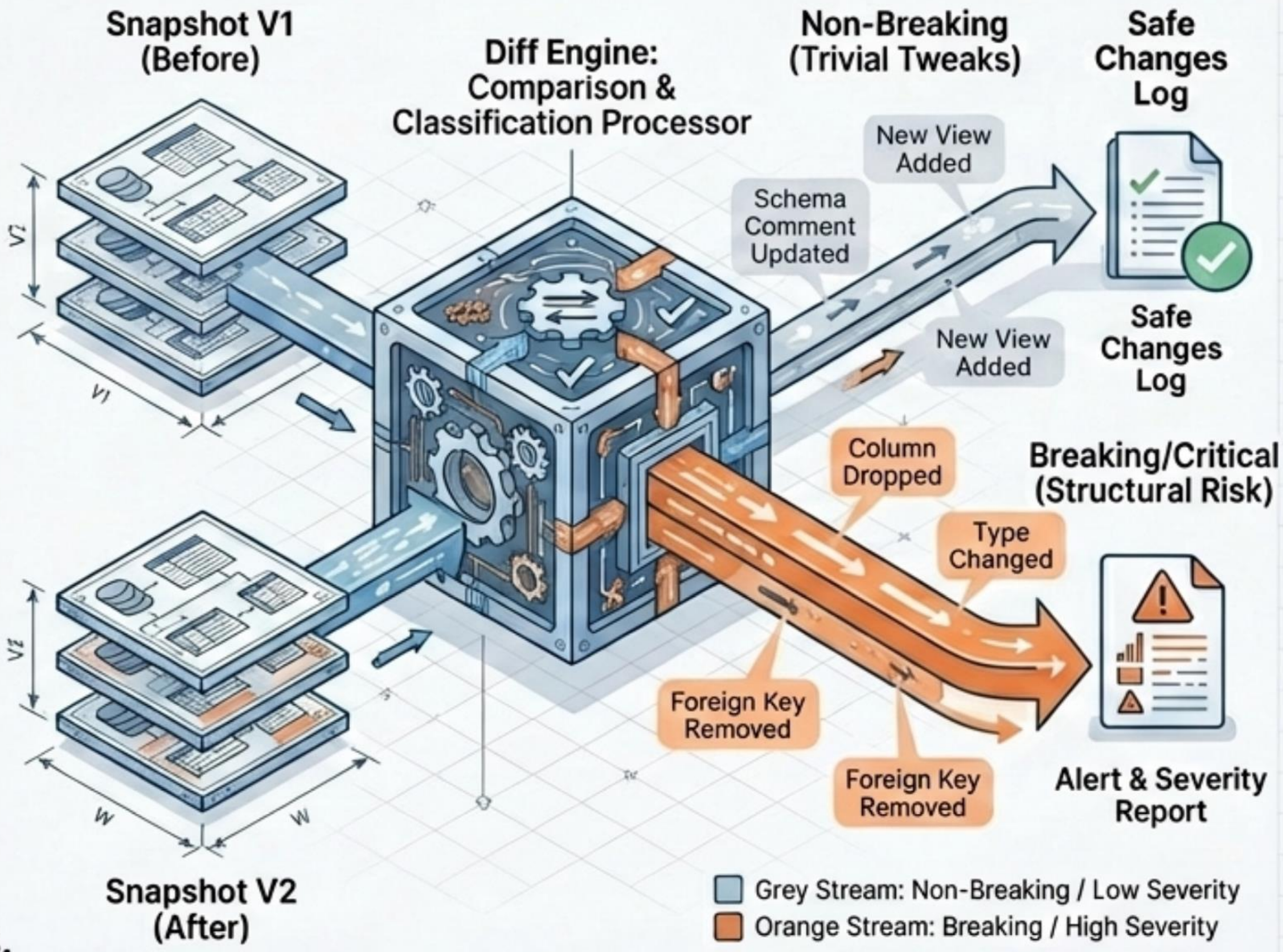
Engine 3: Diff Engine

Role / Purpose:
The Change Detective. Compares snapshots to identify and classify structural evolution.

Core Capabilities:
- Deterministic 'Before/After' comparison.
- 'Breaking' vs. 'Non-Breaking' classification.
- Severity scoring and Change Grouping by domain.

Technical Value:
Consistent processing; produces audit-ready JSON logs of exactly *what* changed. (Using JetBrains Mono for 'JSON')

Strategic Value:
Eliminates 'alert fatigue' by distinguishing between trivial tweaks and critical structural risks.



Engine 4: Graph & Impact Engine

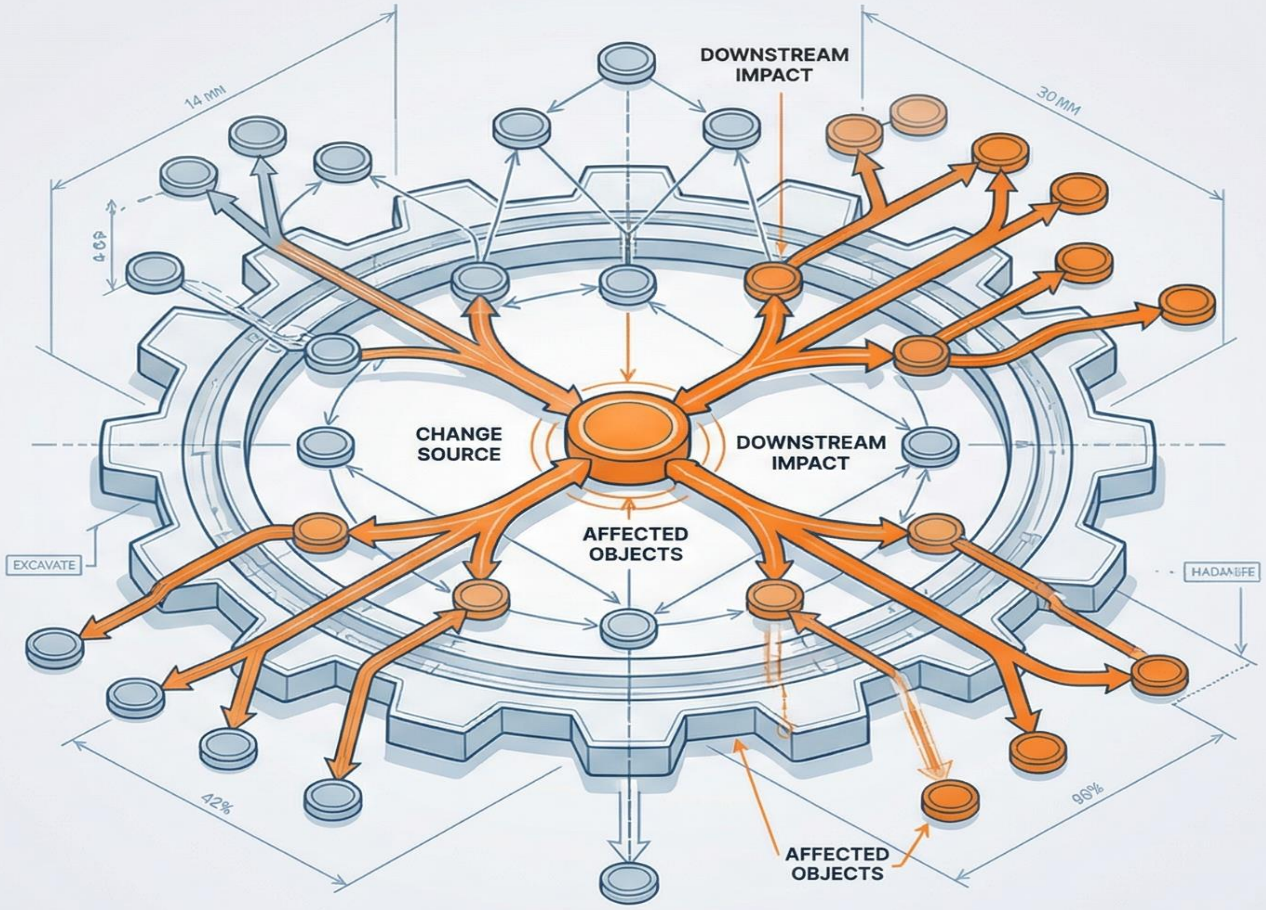
Role / Purpose: The Network Mapper. Consumes *parser-derived lineage* and calculates the “blast radius” of changes.

Core Capabilities:

- Builds versioned dependency graph from *parser-derived lineage* (Nodes & Edges).
- Scores fragility and Centrality metrics.
- Cycle detection and Critical Path analysis.

Technical Value: Native SQL graph logic removes dependency on external graph databases.

Strategic Value: Proactively predicts outages by visualizing downstream impact before deployment.



Engine 5: Usage & Criticality Engine

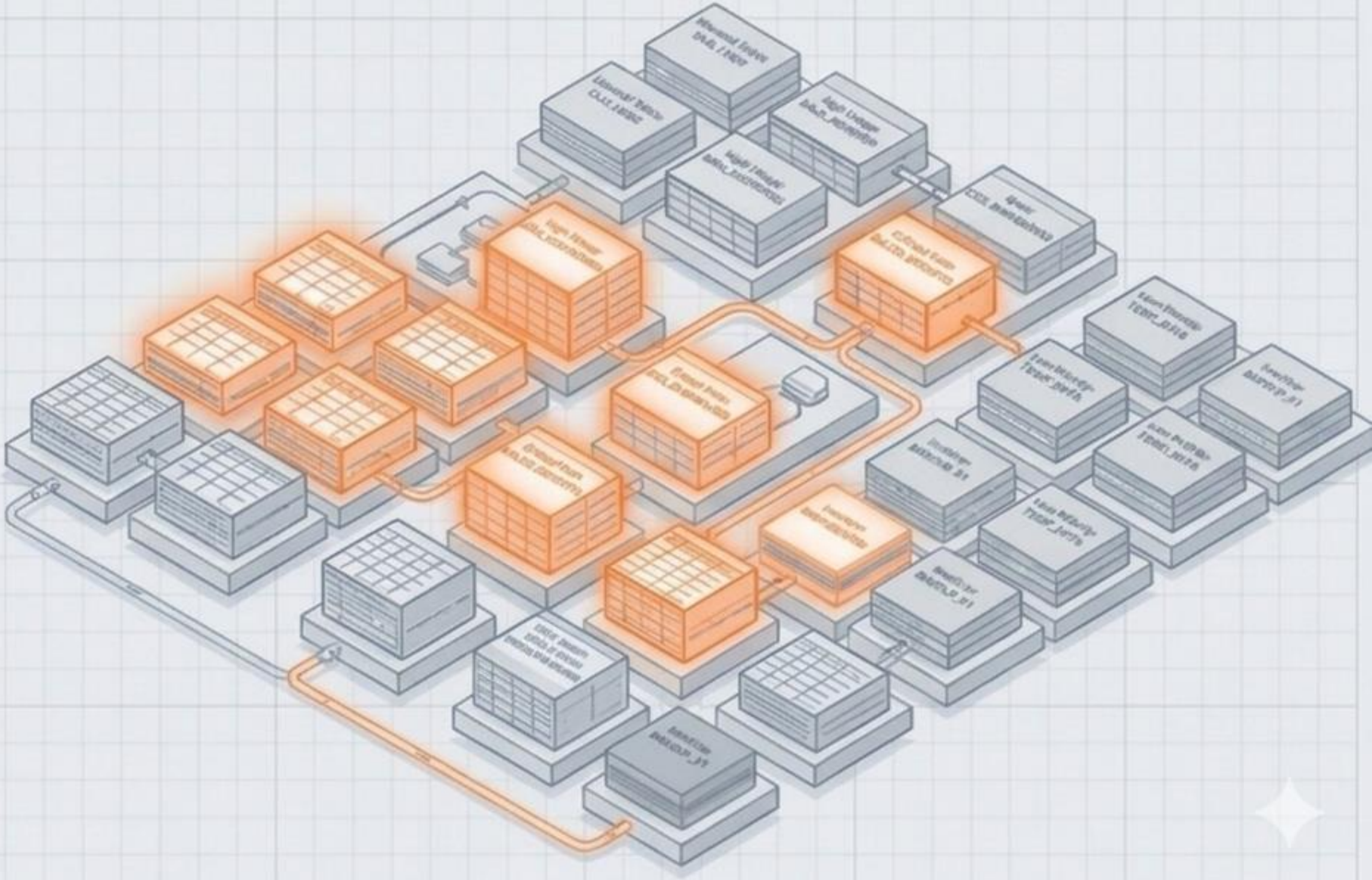
Engineering Transparency

Role / Purpose: The Contextualizer. Weights technical structure with actual business usage reality.

Core Capabilities:
- Query log ingestion and parsing.
- Usage frequency heat-mapping.
- User-weighting (e.g., CEO Report > Ad-hoc query).

Technical Value: Merges static metadata (structure) with dynamic logs (runtime).

Strategic Value: Shifts governance from theoretical impact to actual business risk.



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Engine 6: Intelligence Metrics Layer

Role / Purpose:

The Analyst. Aggregates raw signals into high-level ecosystem health indicators.

Core Capabilities:

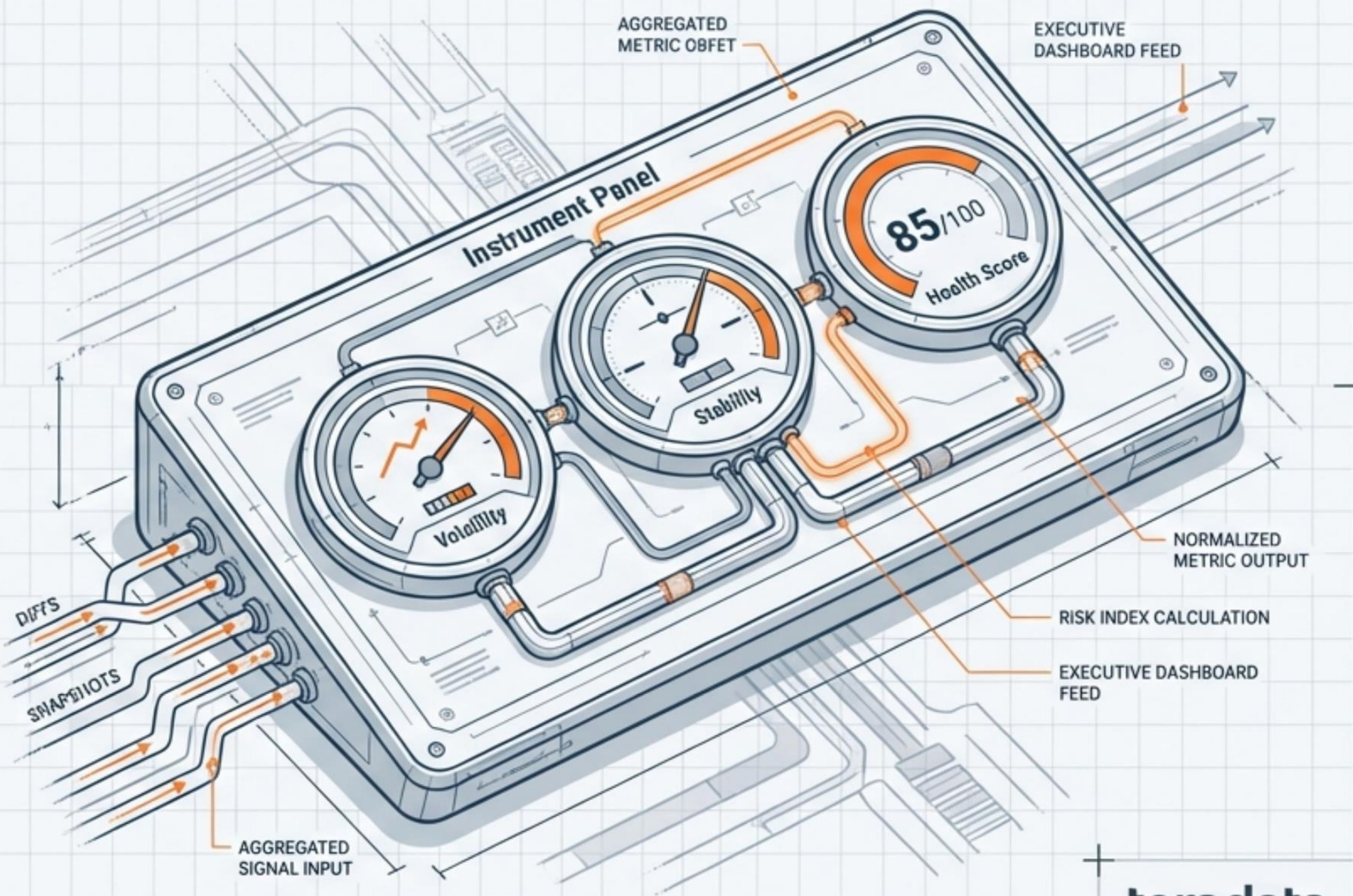
- **Volatility Indexing** (identifying unstable domains). (JetBrains Mono)
- **Stability Trends** per object. (JetBrains Mono)
- **Governance Scorecards** and **Domain Risk Indices.** (JetBrains Mono)

Technical Value:

Transforms disparate data points (diffs, snapshots) into standardized metrics.

Strategic Value:

Elevates SCION from a 'Metadata Tracker' to an 'Enterprise Structural Intelligence Platform'.



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Engine 7: TAISA Integration Layer

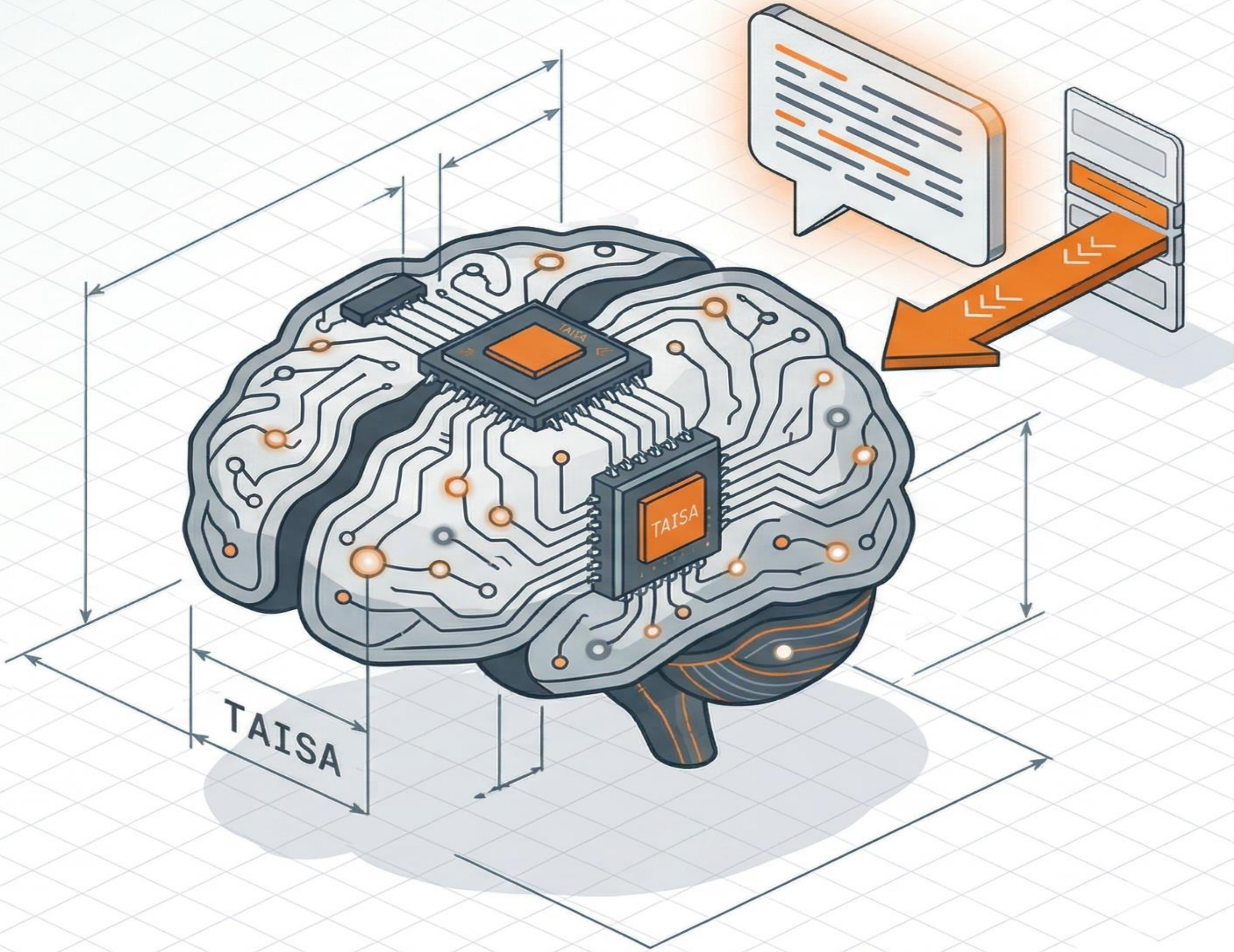
Role / Purpose: The AI Augmentation. Interprets structured change and impact events and explains risk in natural language.

Core Capabilities:

- Consumes *parser-derived lineage* from the platform.
- Structured event interpretation and risk explanation.
- Deterministic context input from versioned system events.

Technical Value: AI operates as “Read-Only”—it interprets facts but cannot corrupt state.

Strategic Value: Deploys internal AI (TAISA) as an always-on expert analyst.



Infrastructure & Deployment Requirements

Specs & Runtime

Minimum On-Prem / VM Requirements:
- Compute: 4–8 vCPU
- Memory: 16–32 GB RAM
- Storage: 200–500 GB (Low footprint)
- Runtime: Docker / Fully Containerized

Cloud Instance Estimate:
- ~$200-$600/month (Region dependent)
- ~$2,400-$7,200/year (Estimated)
- Scales horizontally via K8s/Docker Swarm

Cost Contrast

Zero Licensing Fees. No SaaS Dependency.
100% Owned Infrastructure.

Engineering Transparency Required Team Profile & Development Model

Internal Engineering Initiative | Not a Consulting Engagement

Lead Engineer (1 FTE)

- **Focus:** Architecture, Core Logic (Diff/Graph), State Management
- **Responsibility:** System integrity and critical path analysis

Backend Engineer (1 FTE)

- **Focus:** API Orchestration, Ingestion Adapters, Optimization
- **Responsibility:** Scalability and data flow.

Required Developer Knowledge

- **Programming:** Python, SQL, (Java/Go optional)
- **Architecture:** Microservices, API Design (REST/gRPC), Event-Driven
- **Data & Logic:** Graph Databases, Diff Algorithms, State Management
- **Infrastructure:** Docker, Kubernetes, CI/CD pipelines.

UI Development Plan

Phase	Activity	Output	Tools
Design	Wireframing, UX Flows	Mockups, Prototype	Figma
Development	Frontend Implementation	Responsive Web App	React/Angular, CSS
Integration	Connect to APIs	Functional UI	REST/GraphQL
Testing	UAT, Bug Fixing	Stable UI	Jest, Cypress

Internal Continuity

Development is owned by Teradata DNA. Source code and architectural knowledge are retained internally, eliminating "black-box" dependency.

Delivery Timeline & Execution Scenarios

Scenario A: High Velocity (Recommended) | 2 Engineers | 5-6 Months Timeline

Lead Engineer (1 FTE)

Focus:
- Architecture, Core Logic (Diff/Graph), State Management.

Key Tasks:
- Define system architecture & patterns.
- Implement core diffing & graph logic.
- Manage state consistency.
- Perform critical path analysis.

Backend Engineer (1 FTE)

Focus:
- API Orchestration, Ingestion Adapters, Optimization.

Key Tasks:
- Develop API orchestration layer.
- Build ingestion adapters for data sources.
- Optimize performance & scalability.
- Manage data flow and pipelines.

Milestones:

M2: Shadow Mode MVP (Risk Reduction)

M6: DNA-Ready Release

Delivery Timeline & Execution Scenarios

Scenario B: Low Velocity | 1 Engineer | 9–12 Months Timeline

Full-Stack Engineer (1 FTE)

Focus:
- End-to-end ownership (Architecture + Backend + Integration)
- Incremental delivery & prioritization

Key Tasks:
- Define system architecture & patterns
- Implement core engines (Snapshot, Diff, Graph)
- Build API orchestration layer
- Develop ingestion adapters
- Integrate TAISA reasoning
- Manage performance and scalability

Milestones:
- **M3–M4**: Core Engines MVP (Snapshot + Diff)
- **M6–M8**: Lineage & Impact Ready
- **M9–MI2**: Lineage & Impact Ready
- **M9–M12**: DNA-Ready Release

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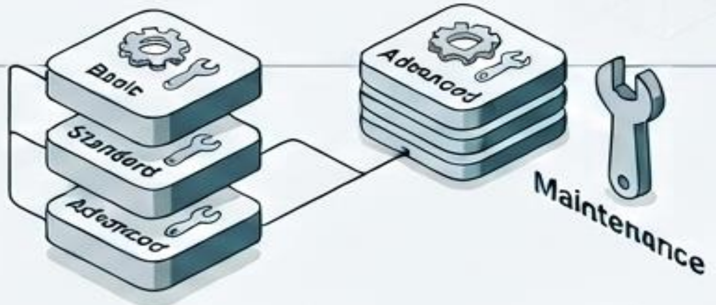
SCION Maintenance Model & Effort Estimation

Flexible maintenance model aligned with enterprise service tiers



Capability	Basic (Run & Secure)	Standard (Sustain & Adapt)	Advanced (Evolve)
Security patches	✓	✓	✓
Dependency updates	✓	✓	✓
Critical bug fixes	✓	✓	✓
Compatibility updates	✗	✓	✓
Minor performance tuning	✗	✓	✓
Environment adaptation	✗	✓	✓
New features / enhancements	✗	✗	✓
TAISA evolution	✗	✗	✓

Maintenance Effort & Tech Profile Estimation (Monthly)

Tier	Estimated Effort	Required Technical Profile
Basic	8-12h hours/month	**Infrastructure Operations Specialist** Monitors system health, applies routine updates, and manages common incidents.
Standard	16-24h hours/month	**Senior Platform Engineer** Optimizes system performance, manages security configurations, and adapts the infrastructure to new requirements.
Advanced	40+h hours/month	**Principal Architect** Defines the technical roadmap, designs complex new features, and leads the strategic evolution of the platform.

Notes

Recommended: Standard Maintenance for production environments
Effort estimates can be translated into cost based on internal resource rates
Estimates assume stabie architecture and no major scope changes
TAISA Integration is non-blocking and independently maintainabie

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SCION – Consolidated View

Integrated overview of architecture, delivery model, and operational footprint

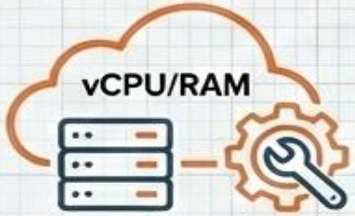
Architecture

- Modular layered design
- Deterministic processing (Snapshot + Diff + Graph)
- AI augmentation via TAISA (read-only, non-invasive)

Estimation

- 2 Engineers (1 Lead + 1 Backend)
- 5–6 months delivery timeline
- Shadow Mode MVP by Month 2

Infrastructure & Maintenance



- Low footprint (4–8 vCPU / 16–32 GB RAM)
- No licensing cost (fully owned platform)
- Flexible maintenance model (Basic / Standard / Advanced)

SCION enables a transition from vendor dependency to fully owned structural intelligence.

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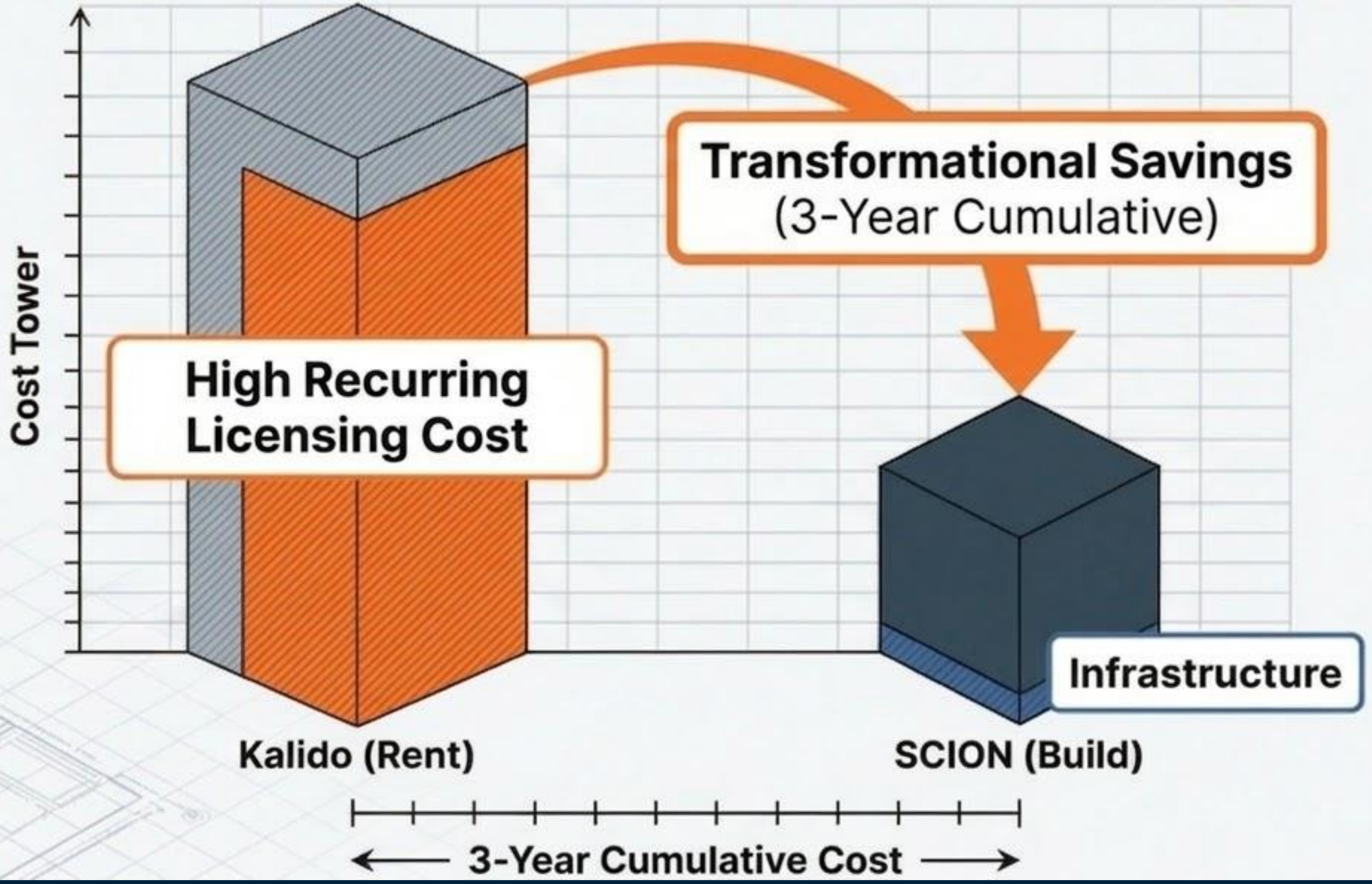
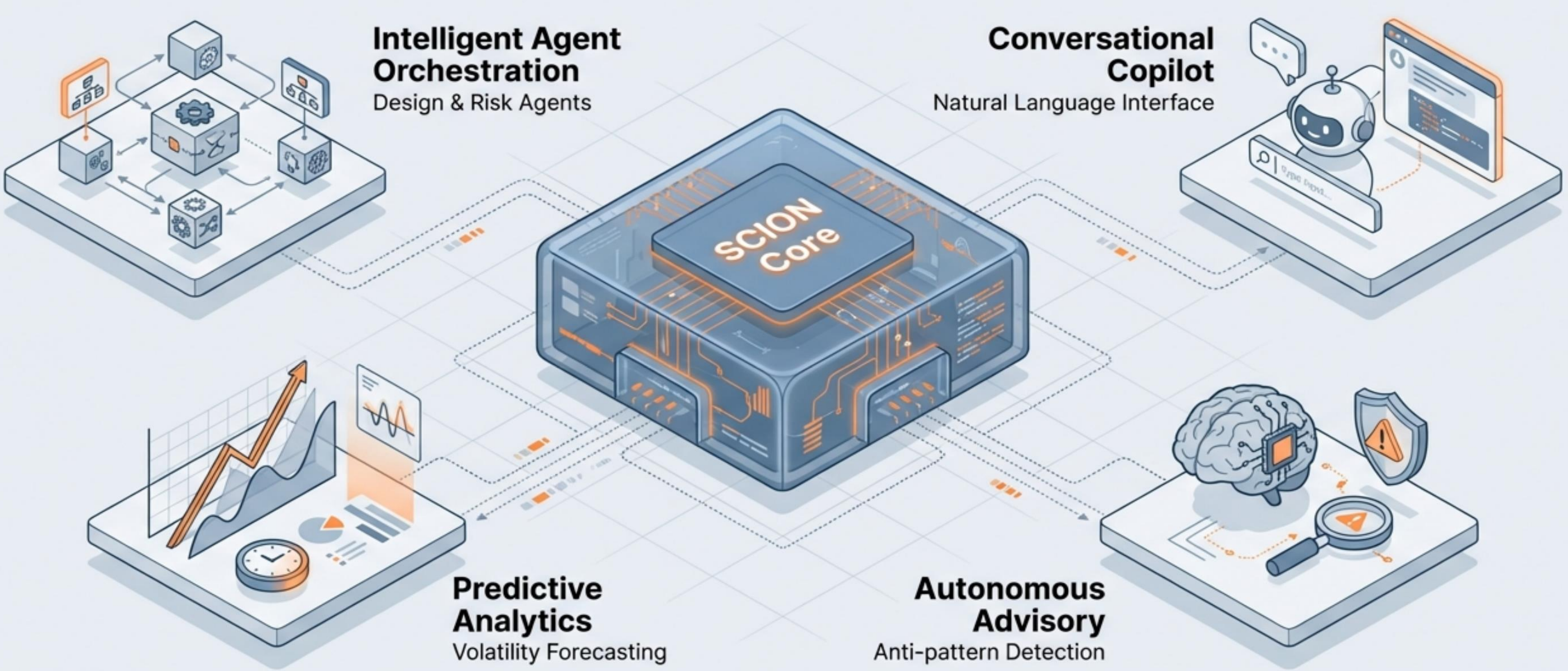
Financial & Strategic Impact



- **Immediate:** Eliminates recurring vendor cost.
- **Asset:** Shifts spend from Renting (OpEx) to Building (IP).
- **Risk:** Eliminates "black-box" dependency.

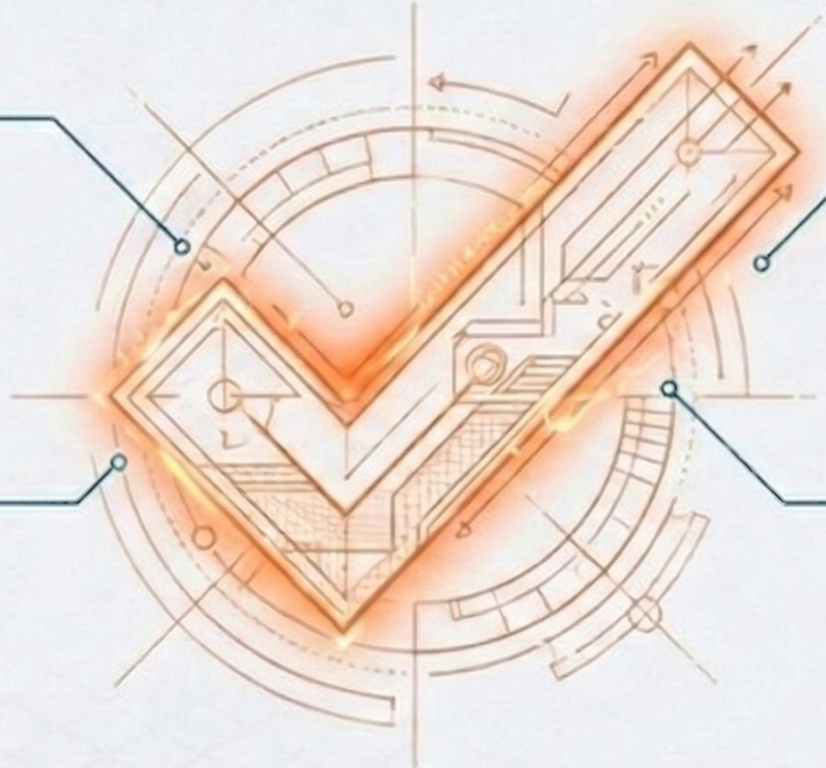
Phase 2 Strategic Potential (Post-DNA Ready)



Executive Recommendation

Financially Justified: Clear ROI and elimination of substantial recurring platform expenses.

Technically Viable: MVP validated Snapshot/Diff logic.



Architecturally Sound: Modular, AI-ready design.

Strategically Essential: Moves DNA to proprietary independence.

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